Features in Development

For CTO review: flexibility and fit with current features

Colosseum · 7 November 2026 · Proposed, not yet built · 19 features in 3 groups

These features sit on top of the Phase-1 design pack (docs 01–13). Both are **learning features**: they help students see and improve what they can do, they **never change a score or a rating**, they read only submitted work, and companies see none of their data in the first version.

What we're asking you to review

1. Flexibility

How much room do we have to build these? Can the **Now** items go in this quarter without slowing the core loop, is **Next** realistic straight after it, and are the deferrable items the right ones to cut first?

2. Fit with current features

Does anything clash with the existing design: module boundaries, the data model, the AI service, company views, or the scoring, rating and integrity rules? The fit matrix on page 2 marks the places to check.

The 19 features

ID	Feature	Timing	Group
A1	Skill Tree	Next	Skill Mastery
A2	Sub-skill Tags on Rubrics	Now	Skill Mastery
A3	Evidence-earned Tiers	Next	Skill Mastery
A4	Mastery Fade and Refresh	Later	Skill Mastery
A5	Traceable Mastery	Next	Skill Mastery
A6	Sub-skill Estimates	Later	Skill Mastery
B1	Assumption Catalogue	Now	Assumption Tracking
B2	Exposure Mapping	Now	Assumption Tracking
B3	Answer Tagging	Next	Assumption Tracking
B4	Assumption Patterns	Later	Assumption Tracking
B5	Your Assumptions Page	Next	Assumption Tracking
B6	Targeted Practice	Later	Assumption Tracking
B7	Tag Disputes	Next	Assumption Tracking
B8	Cohort Assumption Insights	Later	Assumption Tracking
C1	Learning-profile Consent	Next	Shared Foundations
C2	Company Exclusion	Next	Shared Foundations
C3	Data Model Additions	Next	Shared Foundations
C4	Assumption-tagging Endpoint	Next	Shared Foundations
C5	Quality Measurement	Next	Shared Foundations

Now: can start this quarter without touching the core loop. **Next**: first step after the core loop (decision D14). **Later**: once Next is live and has data.

Fit with current features

How each feature group touches what is already designed. Amber cells are worth checking first.

Current feature (Phase-1 doc)	Skill Mastery	Assumption Tracking
Challenge authoring and certifier (05)	Extends: sub-skill tags and one check (A2)	Extends: exposure tags and one check (B2)
AI judge and evidence check (04)	Reuses: criterion scores as they are	Reuses: the quote check, in a separate call (B3)
Rating and decay (04, 07)	Reuses: the same maths at sub-skill level; rating unchanged	No change
Practice attempts (04 §3.2)	Reuses: practice counts toward tiers	Reuses: practice answers are tagged
Instance generation and bank (05 §5)	No change	Needs check: targeted draw in practice mode (B6)
Integrity and activity data (06)	No change: activity data never read	No change: activity data never read
Student scorecard (04, 08)	Extends: links to tiers	Extends: a tag section for the student (B5)
Company search and scorecards (08)	Must exclude (C2)	Must exclude (C2)
College reporting (01 §6)	Extends: aggregates for 25 or more	Extends: cohort insights for 25 or more (B8)
Consent and erasure (06, 08, 09)	Extends: new consent purpose (C1)	Extends: new consent purpose (C1)
Data model (09)	Extends: four new tables	Extends: five new tables
AI service (03)	No change	Extends: one new endpoint (C4)
Module boundaries (01 §4)	Needs check: a new mastery module needs an ADR	Same module

Reuses | Extends | No change | Needs check | Must exclude

Flexibility at a glance

NOW (3)

A2 Sub-skill Tags on Rubrics
B1 Assumption Catalogue
B2 Exposure Mapping

NEXT (11)

A1 Skill Tree
A3 Evidence-earned Tiers
A5 Traceable Mastery
B3 Answer Tagging
B5 Your Assumptions Page
B7 Tag Disputes
C1 Learning-profile Consent
C2 Company Exclusion
C3 Data Model Additions
C4 Assumption-tagging Endpoint
C5 Quality Measurement

LATER (5)

A4 Mastery Fade and Refresh
A6 Sub-skill Estimates
B4 Assumption Patterns
B6 Targeted Practice
B8 Cohort Assumption Insights

Can't be cut (they keep the guarantees)	Can be cut or deferred first
A5 Traceable Mastery · **C1** Learning-profile Consent · **C2** Company Exclusion · **C5** Quality Measurement	**A4** Mastery Fade and Refresh · **A6** Sub-skill Estimates · **B6** Targeted Practice · **B8** Cohort Assumption Insights

Who does the work. B1 is practitioner work and can run in parallel without engineering. A2 and B2 are one column each in template authoring. Everything else is engineering after the core loop. **Model cost:** Skill Mastery none; Assumption Tracking about ₹0.25 per scored answer (est.).

A Skill Mastery (game style)

Shows students, skill by skill, what they have proven with real scored work, in a form that is fun to fill in and honest about skills going stale.

A1 Skill Tree — NEXT

WHAT IT DOES	A tree of about 54 sub-skill nodes across CSE, ECE and IT (18 families × 3 sub-skills). Nodes light up only from scored work, never from signing in, streaks or attempting.
FITS WITH	The domain taxonomy (doc 07) supplies the nodes; the student scorecard links to it.
FLEXIBILITY	Depends on A2 and A3. Can launch as a plain list before the visual tree. Needs frontend and backend time.

A2 Sub-skill Tags on Rubrics — NOW

WHAT IT DOES	Each rubric criterion names the sub-skills it tests. The certifier checks that every certified variant can exercise each tagged sub-skill.
FITS WITH	Extends challenge authoring and the certifier (doc 05). Tags are versioned with the rubric (D2).
FLEXIBILITY	No dependencies. One column plus one certifier check, so it can go in this quarter. Starting now means evidence builds up from the first attempt.

A3 Evidence-earned Tiers — NEXT

WHAT IT DOES	Unlit → Proven (level 3 or higher on one variant) → Mastered (level 3 or higher on two different variants), plus Needs a refresh. Working names and thresholds.
FITS WITH	Reads the criterion scores the judge already produces (doc 04). Ranked and practice attempts both count. Never feeds the rating.
FLEXIBILITY	Depends on A2. Thresholds are configuration, so they can change without a migration.

A4 Mastery Fade and Refresh — LATER

WHAT IT DOES	After the 90-day grace period, tiers fade at each family's decay rate. One new qualifying answer restores them.
FITS WITH	Reuses the rating's decay rates and grace period (docs 04 §4, 07 §4).
FLEXIBILITY	Can be deferred: tiers work without fading at first. Pure computation, no model cost.

A5 Traceable Mastery — NEXT

WHAT IT DOES	Every tier change is an append-only event citing the criterion score, and through it the quote, that earned it.
FITS WITH	Same append-only, evidence-backed pattern as score decisions (docs 01 §8, 09 §4).
FLEXIBILITY	Ships with A3 and should not be cut: it keeps mastery as reconstructable as scores.

A6 Sub-skill Estimates — LATER

WHAT IT DOES	A hidden estimate with uncertainty per node, used to pick useful practice. Not shown as a number.
FITS WITH	Reuses the rating update rule (doc 04 §3) at sub-skill level.
FLEXIBILITY	Optional: tiers work without it. Advanced models wait until each node has thousands of observations.

B Assumption Tracking (cognitive modelling)

Finds the assumptions a student's answers keep relying on that the challenge never guaranteed, shows the evidence, and targets practice at them.

B1 Assumption Catalogue — NOW

WHAT IT DOES	A fixed, signed-off list: about 20 shared assumptions (scale, input validity, time, concurrency, partial failure, silent failure, trust boundaries) and about 10 per family.
FITS WITH	Seeded from the rubric levels in doc 07; version-locked like rubrics (D2).
FLEXIBILITY	Practitioner work with no engineering time, so it can run in parallel now. Natural author: the CSE subject reviewer (doc 13).

B2 Exposure Mapping — NOW

WHAT IT DOES	Each template's variable values are tagged with the assumptions they can expose, for example `failure_mode = out_of_order` exposes "events arrive in order". Patterns are measured as held ÷ exposed.
FITS WITH	Extends template parameter slots and the rubric binding (doc 05 §3.1).
FLEXIBILITY	One column at authoring, alongside A2. Needs a first draft of B1.

B3 Answer Tagging — NEXT

WHAT IT DOES	Each scored answer gets tags marked held, guarded or stated, each with an exact quote. The server refuses any tag whose quote is not in the answer.
FITS WITH	Uses the same quote check as the judge's evidence (doc 04 §1.5), but runs as a separate AI call.
FLEXIBILITY	Depends on B1, B2 and C4. Being separate from the judge means it can be switched off without touching any score.

B4 Assumption Patterns — LATER

WHAT IT DOES	Emerging (held twice), Pattern (held 3 or more times, in at least half the exposures, across 2 or more templates) and Outgrown (guarded on the last 3 exposures).
FITS WITH	Outgrown can light a mastery node (A3).
FLEXIBILITY	Waits until B3 passes the quality bar (C5). Thresholds are configuration.

B5 Your Assumptions Page — NEXT

WHAT IT DOES	The student's view: patterns, the quotes behind them, a suggested practice variant, and what they have outgrown. Wording describes the work, never the person.
FITS WITH	Starts as a section on the existing scorecard; tags can explain why a criterion lost a level without changing the score.
FLEXIBILITY	Can begin as per-answer tags on the scorecard and grow into a full page once B4 is live.

Illustrative B5 card. You tend to assume only one worker touches a record at a time. Held in 3 of 5 answers that could show it, across 3 challenges.
· Webhook challenge: the duplicate check and the insert run in separate transactions
· Settlement worker: balance read and debit with no lock or version check
Try: a practice variant with 16 concurrent workers. **Outgrown so far:** "the result set is never empty" ✓

B6 Targeted Practice

LATER

WHAT IT DOES	In practice mode, the variant is chosen to expose the student's open patterns, for example `workers = 16`.
FITS WITH	Practice attempts (doc 04 §3.2) and instance generation (doc 05 §5). Ranked attempts keep the normal seeded draw.
FLEXIBILITY	Can be deferred. Needs a check against the instance bank and seeding design before building.

B7 Tag Disputes

NEXT

WHAT IT DOES	A "that's not what I assumed" button. The tag is hidden and the dispute is logged as a new row.
FITS WITH	A lighter cousin of score appeals (doc 01 §5.3). Dispute rates feed C5.
FLEXIBILITY	Small; ships with B5.

B8 Cohort Assumption Insights

LATER

WHAT IT DOES	Colleges see the most common assumptions for cohorts of 25 or more, as curriculum feedback.
FITS WITH	Follows the college aggregate-only rule and the 25-student floor (docs 01 §6, 04 §5.3).
FLEXIBILITY	Can be deferred; needs enough students per cohort.

C Shared Foundations

The consent, privacy, data and AI-service pieces both features depend on.

C1 Learning-profile Consent — NEXT

WHAT IT DOES	A new consent purpose, off by default, covering mastery and assumption data. The erasure job covers every new table.
FITS WITH	Extends the versioned consent records (docs 06 §3, 08 §1.3) and the erasure job (doc 09 §1.1).
FLEXIBILITY	Required before any of this data is shown to a student. Not optional.

C2 Company Exclusion — NEXT

WHAT IT DOES	Mastery and assumption data never appear in company search or unlocked scorecards in the first version.
FITS WITH	Enforced in the disclosure code (doc 08 §1.3.1), not by page layout.
FLEXIBILITY	Required, and small. Not optional.

C3 Data Model Additions — NEXT

WHAT IT DOES	Nine tables: `skill_node`, `criterion_skill_tag`, `mastery_event`, `mastery_state`, `assumption`, `item_exposure`, `assumption_observation`, `assumption_dispute`, `assumption_profile`.
FITS WITH	Follows doc 09's append-only and access rules. A new top-level mastery module needs an ADR (doc 03 §5).
FLEXIBILITY	Lands table by table with each feature rather than all at once.

C4 Assumption-tagging Endpoint — NEXT

WHAT IT DOES	`POST /v1/tag-assumptions` on the AI service. Accepts only the assumptions a variant exposes; returns tags with quotes, never free text.
FITS WITH	Same safety rules as the existing endpoints: no database credentials, no storage writes (docs 03 §3, 09 §8).
FLEXIBILITY	Needed for B3. About ₹0.25 per scored answer (est., based on doc 03's integrity classifier price).

C5 Quality Measurement — NEXT

WHAT IT DOES	Tag precision of 0.85 or higher (about 50 tags audited a week), student agreement of 70% or higher, dispute rates, and whether the held rate falls after targeted practice.
FITS WITH	Mirrors the judge's calibration gate (doc 03 §8).
FLEXIBILITY	Required: it gates B4. Needs about an hour or two of practitioner time a week.

Rules both features keep

- They never change a score or a rating.
- They read only submitted work, never typing, paste or focus data.
- Every tier change and every tag is backed by quoted evidence.
- Fixed, signed-off vocabularies; no free-text labels about people.
- No points for participation.
- Companies see nothing in the first version; colleges see aggregates of 25 or more.
- Off by default, under their own consent purpose.

Open decisions

Decision	Recommendation
Tier names and thresholds	Keep the working names until student testing
Show assumption tags on rated scorecards, or only in practice?	Both; never in company views
Should targeted practice use a student's history?	Yes, in practice mode only
Who writes the catalogue?	The CSE subject reviewer (doc 13), signed off by the lead
Should mastery ever be shown to companies?	Not in the first version